

ERZURUM, Turkey

WMO: 170960

Lat: 39.953N

Lon: 41.190E

Elev: 1758

StdP: 81.91

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.1	-25.1	-30.8	0.3	-28.2	-27.4	0.4	-25.0	10.9	-5.6	9.9	-6.2	0.9	220	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.6	31.0	14.6	29.3	14.4	27.9	14.4	16.9	26.6	16.1	25.7	15.3	24.9	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.7	21.0	12.2	11.0	20.2	11.6	10.5	19.6	54.9	26.8	51.9	25.5	49.2	24.7	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.9	8.0	DB	-31.2	33.3	3.8	1.8	-33.9	34.6	-36.1	35.7	-38.3	36.7	-41.0	38.0	(4)
(5)			WB	-31.4	18.8	3.7	1.7	-34.0	20.0	-36.2	20.9	-38.3	21.9	-41.0	23.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-9.9	-8.5	-1.6	5.6	10.5	14.6	18.9	19.4	14.1	8.3	0.7	-6.8
	DBStd	11.07	6.75	6.65	5.70	3.59	2.79	2.58	2.61	2.42	2.96	3.29	4.57	7.11
	HDD10.0	2536	618	517	360	136	28	1	0	0	4	71	278	522
	HDD18.3	4774	876	751	618	382	243	114	23	17	130	312	528	780
	CDD10.0	900	0	0	0	5	43	139	277	291	127	18	0	0
	CDD18.3	97	0	0	0	0	0	3	42	49	3	0	0	0
	CDH23.3	2020	0	0	0	0	4	138	751	917	208	2	0	0
	CDH26.7	598	0	0	0	0	0	14	235	313	36	0	0	0
Wind	WSAvg	3.0	2.2	2.4	3.2	3.6	3.3	3.2	3.7	3.7	3.1	2.8	2.5	2.2
Precipitation	PrecAvg	402	20	23	32	54	69	46	25	17	21	44	32	22
	PrecMax	638	56	90	82	148	154	113	93	63	76	104	81	53
	PrecMin	251	2	1	1	9	15	3	0	0	0	2	0	5
	PrecStd	80	13	15	17	27	30	25	18	16	18	26	17	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	6.0	14.0	20.2	23.8	28.2	33.0	33.1	29.9	23.1	14.2	9.0
		MCWB	1.1	2.4	5.9	9.5	12.5	14.7	15.2	14.6	12.6	10.5	6.2	3.8
	2%	DB	2.4	3.7	10.9	17.6	21.9	26.2	31.0	31.2	27.5	21.0	12.2	5.3
		MCWB	0.3	1.2	4.6	8.7	11.6	13.8	15.0	14.3	12.6	9.9	5.4	2.2
	5%	DB	1.2	2.1	8.7	15.2	20.0	25.0	29.2	30.0	25.8	18.9	10.2	3.8
		MCWB	-0.5	0.2	3.9	7.5	10.6	13.4	14.9	14.2	11.9	9.3	4.6	1.4
	10%	DB	0.0	0.8	6.2	13.0	18.1	23.2	27.8	28.3	24.0	16.6	8.2	2.2
		MCWB	-1.2	-0.7	2.8	6.6	10.2	13.0	14.9	14.3	11.5	8.8	4.0	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.9	6.9	10.9	14.6	17.1	18.7	17.9	15.1	12.3	7.6	4.4
		MCDB	2.9	5.0	13.2	17.5	21.1	24.8	28.1	28.1	25.6	19.7	12.0	7.6
	2%	WB	0.6	1.4	5.3	9.2	12.7	15.4	17.2	16.7	14.0	11.0	6.3	2.6
		MCDB	1.9	3.1	9.7	15.9	19.2	23.8	26.8	26.8	23.9	18.6	10.6	4.5
	5%	WB	-0.3	0.2	4.0	8.1	11.6	14.5	16.4	15.8	13.0	10.0	5.2	1.5
		MCDB	1.0	1.6	7.8	13.9	17.9	22.5	26.0	26.0	23.0	16.6	8.9	3.3
	10%	WB	-1.4	-0.8	2.9	7.1	10.7	13.6	15.6	15.1	12.2	9.1	4.3	0.5
		MCDB	-0.2	0.6	5.8	12.2	16.8	21.4	25.0	25.5	22.0	15.4	7.7	1.9
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.7	10.7	12.0	13.4	15.9	16.8	17.6	18.0	14.8	12.8	10.7
		MCDBR	8.2	9.0	12.9	16.4	17.0	18.8	19.1	19.9	21.3	19.6	16.8	11.3
		MCWBR	6.6	6.9	7.6	9.0	8.7	8.6	7.6	7.7	9.6	10.1	10.5	8.1
	5% WB	MCDBR	7.5	8.3	11.1	14.6	14.8	16.6	16.7	17.6	18.5	16.3	14.0	9.4
		MCWBR	6.3	6.4	6.8	8.5	8.2	8.5	7.6	7.8	8.9	9.0	9.1	7.0
Clear-Sky Solar Irradiance	taub	0.274	0.312	0.367	0.424	0.392	0.356	0.369	0.373	0.322	0.350	0.298	0.270	
	taud	2.233	2.103	2.048	2.011	2.201	2.354	2.336	2.303	2.430	2.286	2.391	2.328	
	Ebn at Noon	887	893	877	849	886	918	901	889	921	844	859	868	
	Edh at Noon	96	128	152	169	143	123	124	125	102	106	82	81	
All-Sky Solar Radiation	RadAvg	2.47	3.61	4.60	4.87	5.68	7.05	7.21	6.45	5.31	3.45	2.36	1.97	
	RadStd	0.26	0.30	0.45	0.51	0.43	0.40	0.39	0.28	0.32	0.28	0.32	0.31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.75	-0.57	N/A	N/A	N/A	+73	+27
(47) Regional (6 neighbors)	N/A	N/A	N/A	+0.64	-0.45	N/A	N/A	N/A	+89	+37

Nomenclature: See separate page